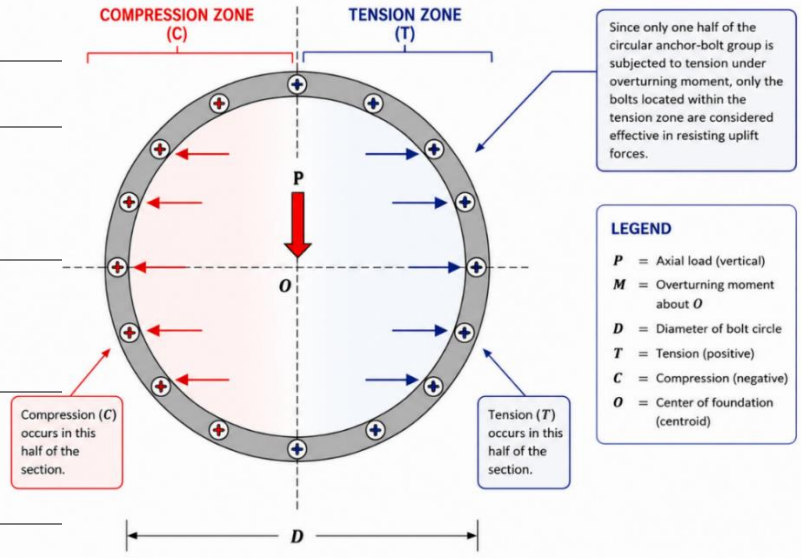
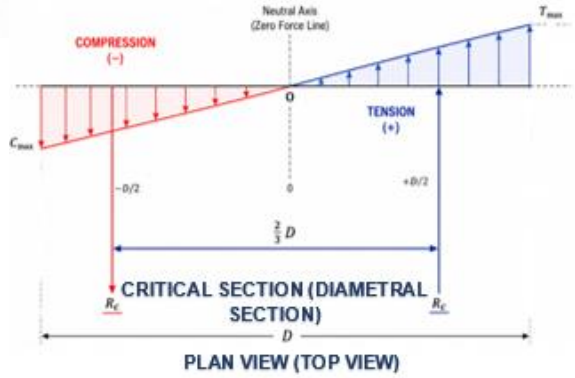
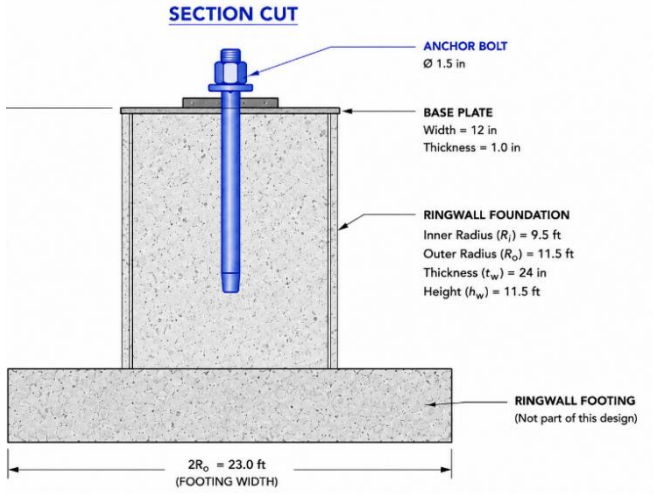
1. Anchor Bolts segment

Design Parameter	Symbol	Value	Source
Factored axial Load (LRFD)	P_u	7607.4940	Main Tank Design (Predefined)
Factored moment (LRFD)	M_u	4976.0383	Main Tank Design (Predefined)
Factored shear force (LRFD	V_u	77.0326	Main Tank Design (Predefined)
Effective embedment depth	Hef	35 inches	User Input
Anchor Bolt Quantity	Nb	16	Assumed to check
Anchor Bolt Diameter	db	1.50 in	Assumed to check
Hole Diameter	dh	1.625 in	ACI Standard Hole
Diameter circle representing axisymmetric	Db	30'-00"	Main Tank Design (Predefined)
Bolt Circle Radius	rb	15.25 ft (183 in)	Calculated
Concrete Strength	f'c	4ksii	User Input
Steel Yield Strength	Fy	60 ksi	User Input
Ultimate Strength	Fu	58 ksi (or project value)	From Code
Resistance Factor	ϕ_t, ϕ_v	0.75/0.75	ACI 318







Ringwall Geometry

From the structural drawings:

Outer Ringwall Diameter = 30 ft

Ringwall Width = 1.667 ft (20 in)

Therefore,

Inner Radius

$$R_i = 15 - \frac{20}{12} = 13.33 \text{ ft}$$

Outer Radius

$$\boldsymbol{R_o = 15.00 \text{ ft}}$$

Anchor Bolt Circle

From the foundation plan, there are $\boldsymbol{N_b = 16}$ **anchor bolts**

Bolt Diameter (in.)	Minimum Embedment (in.)
1.25	**48.91**
1.50	**60.09**
1.75	**75.00**
2.00	**90.32**

2. Governing Overturning Moment

The anchor bolt design is governed by the maximum overturning moment acting at the top of the foundation.

Step 1. Governing Overturning Moment

From the "Loads on the Foundation" calculations:

- **Wind Moment at Top of Foundation**

$$M_w = 6117.08 \text{ ft} - \text{kip}$$

- **Seismic Moment at Top of Foundation**

$$M_e = 5709.21 \text{ ft} - \text{kip}$$

Since

$$6117.08 > 5709.21$$

The governing load case is **Wind**.

Therefore,

$$\text{using limit state} = 0.6(6117.08) = M_u = 4976.0383 \text{ ft} - \text{kip}$$

3. Tension Force Per Bolt

The overturning moment acting on the tank foundation produces uplift in one-half of the anchor bolt group and compression in the opposite half.

For a uniformly distributed circular anchor bolt group, the total uplift force is

$$T_{total} = \frac{M_u}{r_b}$$

where

- M_u= Governing overturning moment
- r_b= Bolt circle radius

Step 2. Bolt Circle Radius

From the structural drawing

$$D_b = 30' - 6"$$

Therefore

$$r_b = \frac{30.5}{2} = 15.25 \text{ ft}$$

Step 3. Total Uplift Force

$$T_{total} = \frac{4976.0383}{15.25}$$
$$T_{total} = 311.01 \text{ kips}$$

Step 4. Tension Per Bolt

There are

$$N_b = 16$$

anchor bolts equally spaced.

Only approximately one-half of the anchor bolts resist uplift.

Therefore

$$N_t = \frac{16}{2} = 8$$

Hence,

$$T_{bolt} = \frac{311.01}{8}$$
$$T_{bolt} = 38.86 \text{ kips}$$

Step 5. Hole Area

The anchor bolt hole area is calculated as

$$A_h = \frac{\pi d_h^2}{4}$$

For a 1.5-in anchor bolt, the standard hole diameter is

$$d_h = 1.625 \text{ in}$$

Therefore,

$$A_h = \frac{\pi(1.625)^2}{4}$$
$$A_h = 2.074 \text{ in}^2$$

Step 6. Angular Spacing

Number of anchor bolts

$$N_b = 16$$

Angular spacing between adjacent bolts

$$\Delta\theta_b = \frac{360}{N_b}$$
$$= \frac{360}{16}$$
$$\Delta\theta_b = 22.5^\circ$$

Step 7. Bolt Coordinates

The anchor bolt circle radius is

$$r_b = 15' - 3"$$
$$r_b = 15.25 \text{ ft}$$

For bolt i

$$\theta_i = (i-1) \times 22.5^\circ$$
$$x_i = r_b \cos\theta_i$$
$$y_i = r_b \sin\theta_i$$

Example

Bolt 1

$$\theta_1 = 0^\circ$$
$$x_1 = 15.25\cos 0^\circ = 15.25 \text{ ft}$$
$$y_1 = 15.25\sin 0^\circ = 0$$

Bolt 2

$$\theta_2 = 22.5^\circ$$
$$x_2 = 15.25\cos 22.5^\circ$$
$$x_2 = 14.09 \text{ ft}$$
$$y_2 = 15.25\sin 22.5^\circ$$
$$y_2 = 5.84 \text{ ft}$$

Bolt	Angle (°)	x (ft)	y (ft)	Location
1	0	15.25	0.00	Right
2	22.5	14.09	5.84	Upper-right
3	45	10.78	10.78	Upper-right
4	67.5	5.84	14.09	Upper-right
5	90	0.00	15.25	Top
6	112.5	-5.84	14.09	Upper-left
7	135	-10.78	10.78	Upper-left
8	157.5	-14.09	5.84	Upper-left
9	180	-15.25	0.00	Left
10	202.5	-14.09	-5.84	Lower-left
11	225	-10.78	-10.78	Lower-left
12	247.5	-5.84	-14.09	Lower-left
13	270	0.00	-15.25	Bottom
14	292.5	5.84	-14.09	Lower-right
15	315	10.78	-10.78	Lower-right
16	337.5	14.09	-5.84	Lower-right

Step 8. Arc Spacing

Convert radius to inches

$$r_b = 15.25 \times 12$$
$$r_b = 183 \text{ in}$$

Arc spacing

$$S_{arc} = r_b \Delta\theta_b \frac{\pi}{180}$$

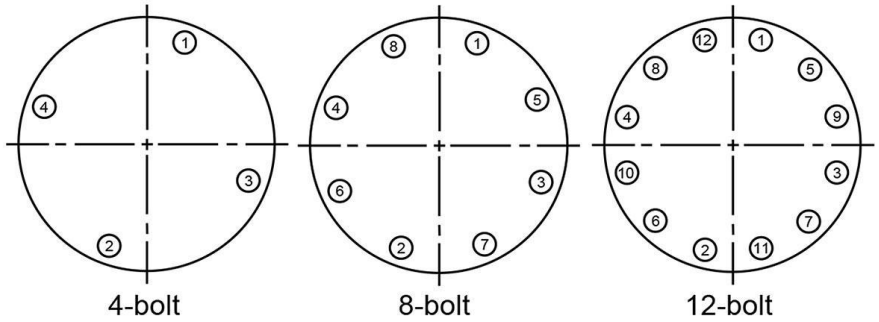
$$S_{arc} = 183(22.5)\frac{\pi}{180}$$
$$S_{arc} = 71.86 \text{ in}$$

Step 9. Chord Spacing

$$S_{chord} = 2r_b \sin\left(\frac{\Delta\theta_b}{2}\right)$$
$$S_{chord} = 2(183)\sin(11.25°)$$
$$S_{chord} = 71.42 \text{ in}$$

Step 10. Number of Bolts Per Segment

$$n_{bs} = \frac{N_b}{n_s}$$

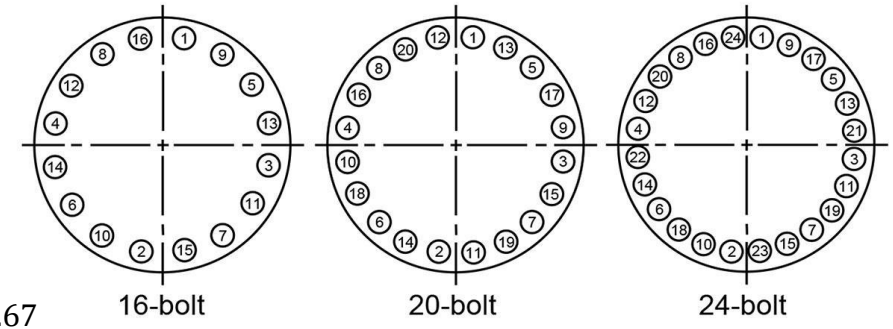


Typical choices

- 4 segments
- 6 segments
- 8 segments

Case 1: 6 Segments

$$n_{bs} = \frac{16}{6} = 2.67$$



Not Acceptable

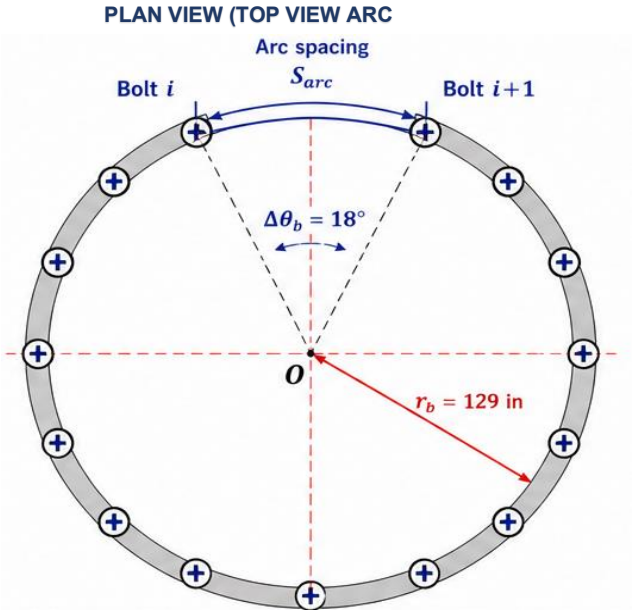
Case 2: 4 Segments

$$n_{bs} = \frac{16}{4} = 4$$

✓ Acceptable

Case 3: 8 Segments

$$n_{bs} = \frac{16}{8} = 2$$



✓ Acceptable

Step 11. Bolt Clear Edge Distance

From the drawing

Base ring width

$$= 12 \text{ in}$$

Bolt circle diameter

$$= 30' - 6"$$

The **radial position of the bolt center within the 12-in base ring** is **not dimensioned** in the drawing. Therefore, the edge distance **cannot be calculated exactly** from the available information.

$$e = 3.0 \text{ in} \ldots. \text{ Assumed}$$

giving

$$e_{clear} = 3.0 - \frac{1.625}{2}$$
$$= 2.19 \text{ in}$$

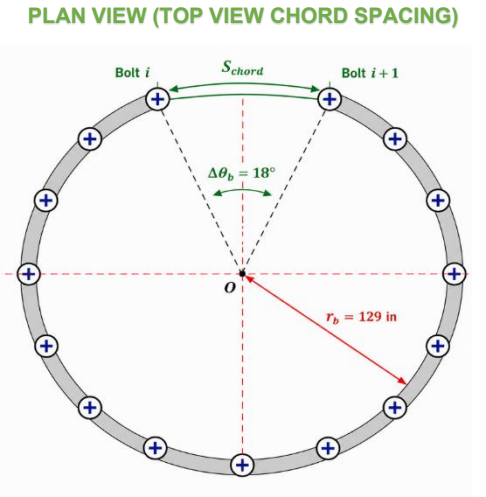
Step 12. Shear Per Bolt

Wind shear

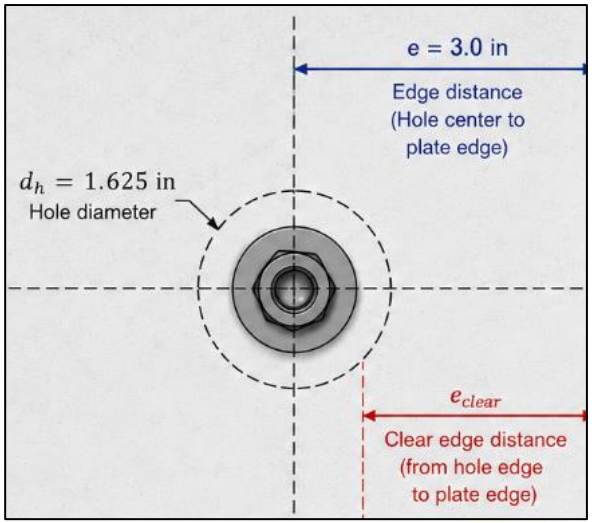
$$V = 77.0326\text{kips}$$

Assuming equal distribution

$$V_{bolt} = \frac{77.0326}{16}$$
$$V_{bolt} = 4.82 \text{ kips}$$



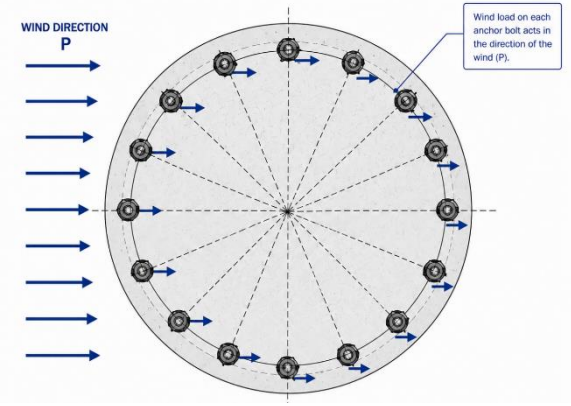
Recommend: Increase edge distance to **3.25–3.5 in**



12. Tensile Capacity

Using:

$$\boldsymbol{\phi P_{nt} = \phi A_b F_u}$$



Given:

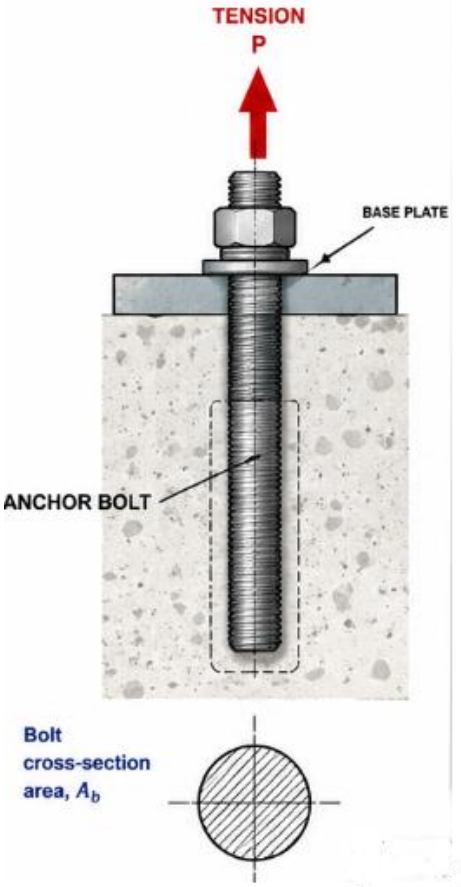
$\phi = 0.75$

$A_b = 1.767$

$F_u = 58$ ksi

$$\phi P_{nt} = 0.75(1.767)(58)$$

$$\boldsymbol{\phi P_{nt} = 76.87 \text{ kips}}$$



13. Shear Capacity

$$F_{nv} = 0.6F_u$$

$$F_{nv} = 0.6(58) = 34.8 \text{ ksi}$$

$$\phi V_n = 0.75(1.767)(34.8)$$

$$\boldsymbol{\phi V_n = 46.11 \text{ kips}}$$

14. Interaction Check

$$\text{Utilization} = \left(\frac{Nu}{\phi N_n}\right)^{\alpha} + \frac{V_u}{\phi V_n} \leq 1 \text{.....} ACI\ 318\text{-}19\ Sec17.6$$

$$\text{Utilization} = \frac{T_{bolt}}{\phi P_{nt}} + \frac{V_{bolt}}{\phi V_n} \leq 1$$

$$= \frac{34.45}{76.87} + \frac{1.034}{46.11}$$

$$= 0.448 + 0.022$$

$$= 0.470$$

$$\boldsymbol{Utilization = 0.47 < 1.0 \Rightarrow SAFE}$$

15. Anchor Bolt Length Design:

- **Minimum embedment (code-based)**

ACI / AISC / industry:

$$h_{ef} \geq 8d_b$$

$$h_{ef} = 8 \times 1.5 = 12\ in$$

16. Pull Out Test/ Concrete Breakout Capacity

$$N_b = k_c\sqrt{f_c'}\ h_{ef}^{1.5}$$

where:

- N_b *= concrete breakout strength*
- k_c *= empirical constant*
- f_c' *= concrete compressive strength*
- h_{ef} *= effective embedment depth.*
- *Embedment = approximately 9–10 ft*
- *Effective embedment = assume* $h_{ef} = 108$ *in conservative*

Based on the foundation detail drawing,

$$h_{ef} = 96\ in$$

Substituting:

$$N_b = 24\sqrt{4000}(96)^{1.5} = 24(63.25)(940.6)$$
$$N_b = 1{,}427{,}470\ lb$$
$$N_b \approx 1427\ kips$$

Using the ACI strength reduction factor:

$$\phi = 0.75$$
$$\phi N_b = 0.75(1427)$$
$$\phi N_b \approx 1070\ kips$$

Maximum tensile demand per anchor bolt:

$$T_{bolt} = 34.45\ kips$$

17. Concrete breakout utilization:

$$Utilization = \frac{T_{bolt}}{\phi N_b}$$
$$= \frac{34.45}{1070}$$
$$= 0.032$$
$$0.032 < 1.0$$

•

Bolt-group tension and shear

For a circular group of equally spaced bolts subjected to a moment about one horizontal axis, a preliminary elastic distribution gives the maximum tension at the extreme bolt as

$$T_{\max} = \frac{2M_u}{N_b r_b}$$

This assumes a rigid base plate, elastic bolt response, and a suitable model for the compression resultant. The actual base-plate/contact model may change the forces.

Using M_u=4,976.0383 kip-ft and r_b=15 ft: $V_{\text{bolt}} = \frac{V_u}{N_b}$

Calculation	16 bolts	20 bolts
Maximum preliminary bolt tension	41.47 kips	33.17 kips
Average shear per bolt	4.81 kips	3.85 kips
Angular spacing	22.5°	18.0°
Chord spacing on 30-ft circle	70.23 in	56.32 in

Preliminary anchor-steel checks

The gross shank area of a 1.50-in bolt is

$$A_g = \frac{\pi(1.50)^2}{4} = 1.767 \text{ in}^2$$

Using gross area and the supplied Fu=58 ksi gives the following illustrative strengths:

$$\phi T_n = 0.75 A_g F_u = 76.87 \text{ kips}$$
$$\phi V_n = 0.75 A_g (0.6 F_u) = 46.12 \text{ kips}$$